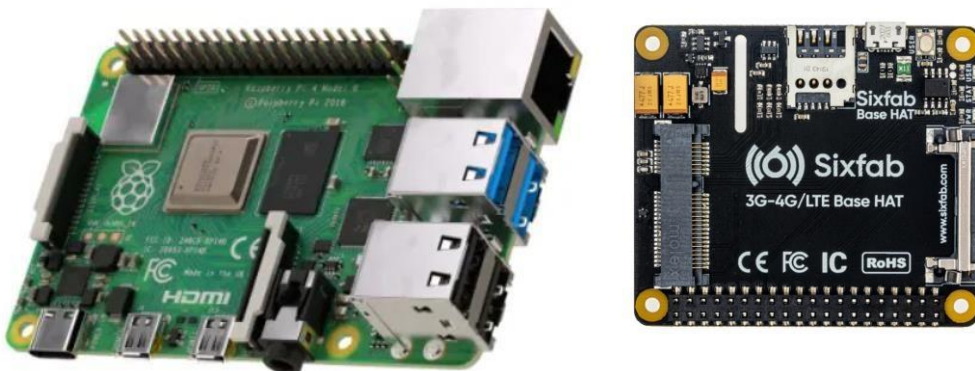


Raspberry pi 4G Connection Setup

In order to establish the connection, we first need a suitable version of rasberian, the operating system in the rpi we are using. The equipment used in this project strangely has compatibility problems with the current 64-bit version. For this reason, the 32-bit legacy version was preferred.

Materials used

- Raspberry pi 8gb
- Sixfab 3G-4G/LTE Base HAT
- Quectel EC25-EUX mini PCL module
- Sim card
- 2 antennas
- SMA cable and fittings



Raspberry pi Sixfab 3G-4G/LTE Base HAT



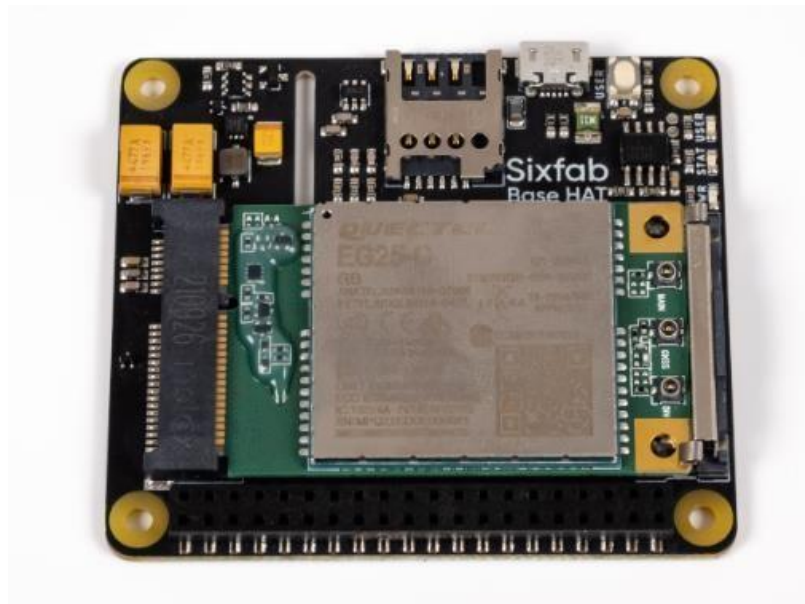
Quectel EC25-EUX mini PCL module SIM card



LTE 12 dpi antenna Sma patch cable

Physical Installation Stages

1. Insert the Quectel EC25/EG25 mini PCIe module into the HAT



2. Attach the antenna to the mini PCIe module.



3. Simply plug the 40-pin header into the Base HAT that comes with the HAT, and the Sixfab Connect SIM is inserted into the Base HAT.



4. Now plug the HAT into the Raspberry Pi.



5. Finally, connect the micro-USB cable to the HAT and Raspberry Pi.



6. The drive is controlled as follows

```
pi@raspberrypi:~$ lsusb
Bus 001 Device 009: ID 2c7c:0125 Quectel Wireless Solutions Co., Ltd. EC25 LTE modem
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 002: ID 2109:3431 VIA Labs, Inc. Hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

The mini PCL module should be visible

Software Installation Stages

1. During the installation of `ppp_install_standalone.sh` Standalone, all source files are downloaded from the internet. Simply download and run it.

```
hasan@raspberrypi:~ $ wget https://raw.githubusercontent.com/sixfab/Sixfab_PPP_Installer/master/ppp_install_standalone.sh
sudo chmod +x ppp_install_standalone.sh
sudo ./ppp_install_standalone.sh
--2024-07-14 21:58:44-- https://raw.githubusercontent.com/sixfab/Sixfab_PPP_Installer/master/ppp_install_standalone.sh
raw.githubusercontent.com (raw.githubusercontent.com) çözümleniyor... 185.199.109.133, 185.199.111.133, 185.199.110.133, ...
raw.githubusercontent.com (raw.githubusercontent.com) [185.199.109.133]:443 bağlanılıyor... bağlantı kuruldu.
HTTP isteği gönderildi, yanıt bekleniyor... 200 OK
Uzunluk: 11075 (11K) [text/plain]
Kayıt yeri: `ppp_install_standalone.sh'

ppp_install_standal 100%[=====>] 10,82K --.-KB/s içinde 0,005s

2024-07-14 21:58:44 (1,98 MB/s) - `ppp_install_standalone.sh' kaydedildi [11075/11075]

2024-07-14 21:59:02 (4,95 MB/s) - `chat-connect' kaydedildi [250/250]

--2024-07-14 21:59:02-- https://raw.githubusercontent.com/sixfab/Sixfab_PPP_Installer/master/src/chat-disconnect
raw.githubusercontent.com (raw.githubusercontent.com) çözümleniyor... 185.199.109.133, 185.199.111.133, 185.199.110.133, ...
raw.githubusercontent.com (raw.githubusercontent.com) [185.199.109.133]:443 bağlanılıyor... bağlantı kuruldu.
HTTP isteği gönderildi, yanıt bekleniyor... 200 OK
Uzunluk: 148 [text/plain]
Kayıt yeri: `chat-disconnect'

chat-disconnect 100%[=====>] 148 --.-KB/s içinde 0s

2024-07-14 21:59:03 (5,54 MB/s) - `chat-disconnect' kaydedildi [148/148]

--2024-07-14 21:59:03-- https://raw.githubusercontent.com/sixfab/Sixfab_PPP_Installer/master/src/provider
raw.githubusercontent.com (raw.githubusercontent.com) çözümleniyor... 185.199.109.133, 185.199.111.133, 185.199.110.133, ...
raw.githubusercontent.com (raw.githubusercontent.com) [185.199.109.133]:443 bağlanılıyor... bağlantı kuruldu.
HTTP isteği gönderildi, yanıt bekleniyor... 200 OK
Uzunluk: 801 [text/plain]
Kayıt yeri: `provider'

provider 100%[=====>] 801 --.-KB/s içinde 0s

2024-07-14 21:59:03 (13,5 MB/s) - `provider' kaydedildi [801/801]
```

Note: this application is the modem application shared on the official site of sixfab company. The 32-bit legacy version was preferred because it caused problems in newly released 64-bit operating systems.

2. Please select your Sixfab Shields/HATs.

You will be asked to choose from the mentioned shields/HATs. Then the necessary scripts will be fetched for you.

Please choose your Sixfab Shield/HAT:

- 1: GSM/GPRS Shield
- 2: 3G, 4G/LTE Base Shield
- 3: Cellular IoT App Shield
- 4: Cellular IoT HAT
- 5: Tracker HAT
- 6: 3G/4G Base HAT

2

```
You chose Base Shield
Checking requirements...
Updating headers...
Ayn1: 1 http://deb.debian.org/debian bookworm InRelease
İndir: 2 http://archive.raspberrypi.com/debian bookworm InRelease [23,6 kB]
İndir: 3 http://deb.debian.org/debian-security bookworm-security InRelease [48,0 kB]
İndir: 4 http://deb.debian.org/debian bookworm-updates InRelease [55,4 kB]
İndir: 5 http://deb.debian.org/debian-security bookworm-security/main armhf Packages [162 kB]
İndir: 6 http://deb.debian.org/debian-security bookworm-security/main arm64 Packages [165 kB]
İndir: 7 http://archive.raspberrypi.com/debian bookworm/main arm64 Packages [418 kB]
İndir: 8 http://deb.debian.org/debian-security bookworm-security/main Translation-en [101 kB]
İndir: 9 http://archive.raspberrypi.com/debian bookworm/main armhf Packages [419 kB]
2 sn.'de 1.288 kB alındı (614 kB/s)
Paket listeleri okunuyor... Bitti
Downloading chat scripts...
--2024-07-14 21:59:02-- https://raw.githubusercontent.com/sixfab/Sixfab_PPP_Installer/master/src/chat-connect
raw.githubusercontent.com (raw.githubusercontent.com) çözümleniyor... 185.199.109.133, 185.199.111.133, 185.199.110.133, ...
raw.githubusercontent.com (raw.githubusercontent.com) [185.199.109.133]:443 bağlanılıyor... bağlantı kuruldu.
HTTP isteği gönderildi, yanıt bekleniyor... 200 OK
Uzunluk: 250 [text/plain]
Kayıt yeri: 'chat-connect'

chat-connect      100%[=====>]      250  --.-KB/s   içinde 0s

2024-07-14 21:59:02 (4,95 MB/s) - 'chat-connect' kaydedildi [250/250]

--2024-07-14 21:59:02-- https://raw.githubusercontent.com/sixfab/Sixfab_PPP_Installer/master/src/chat-disconnect
raw.githubusercontent.com (raw.githubusercontent.com) çözümleniyor... 185.199.109.133, 185.199.111.133, 185.199.110.133, ...
raw.githubusercontent.com (raw.githubusercontent.com) [185.199.109.133]:443 bağlanılıyor... bağlantı kuruldu.
HTTP isteği gönderildi, yanıt bekleniyor... 200 OK
Uzunluk: 148 [text/plain]
Kayıt yeri: 'chat-disconnect'

chat-disconnect   100%[=====>]      148  --.-KB/s   içinde 0s

2024-07-14 21:59:03 (5,54 MB/s) - 'chat-disconnect' kaydedildi [148/148]
```

3. What is your carrier's APN?

It will then ask for the APN. Type in your APN and then press ENTER.

The Carrier APN is exclusive to operators. Since my sim card is bimcell, "internet"

```
What is your carrier APN?
internet
Your Input is : internet
```

4.Does your carrier need a username and password?

Prompts if a user is required for the connection

```
Does your carrier need username and password? [Y/n]
n
You chose n
```

5. Enter your contact PORT name.

It asks for the usb location where you plugged the HAT.

```
What is your device communication PORT? (ttyS0/ttyUSB3/etc.)
ttyUSB3
Your input is: ttyUSB3
```

6. Automatic connection when Raspberry Pi starts?

n was chosen because we wanted to test it.

```
Do you want to activate auto connect/reconnect service at R.Pi boot up? [Y/n]
n
You chose n
```

7. Restart

We press Enter and restart the device.

```
To connect to internet run "sudo pon" and to disconnect run "sudo poff"
Press ENTER key to reboot
```

8. Installation completed. Test time

What needs to be done for the test is to open the terminal and use the sudo pon command.


```

transferpi@raspberrypi:~$ sudo pon
pppd options in effect:
debug                # (from /etc/ppp/peers/provider)
updetach             # (from /etc/ppp/peers/provider)
persist             # (from /etc/ppp/peers/provider)
dump                # (from /etc/ppp/peers/provider)
noauth              # (from /etc/ppp/peers/provider)
remotename 3gppp     # (from /etc/ppp/peers/provider)
/dev/ttyUSB3        # (from /etc/ppp/peers/provider)
115200              # (from /etc/ppp/peers/provider)
lock                # (from /etc/ppp/peers/provider)
connect chat -s -v -f /etc/chatscripts/chat-connect -T bimcell # (from /etc/ppp/peers/provider)
disconnect chat -s -v -f /etc/chatscripts/chat-disconnect      # (from /etc/ppp/peers/provider)
nocrtscts           # (from /etc/ppp/peers/provider)
modem               # (from /etc/ppp/peers/provider)
asynmap 0           # (from /etc/ppp/options)
lcp-echo-failure 4   # (from /etc/ppp/options)
lcp-echo-interval 30 # (from /etc/ppp/options)
hide-password       # (from /etc/ppp/peers/provider)
novj                # (from /etc/ppp/peers/provider)
novjccomp           # (from /etc/ppp/peers/provider)
ipcp-accept-local   # (from /etc/ppp/peers/provider)
ipcp-accept-remote  # (from /etc/ppp/peers/provider)
ipparam 3gppp       # (from /etc/ppp/peers/provider)
noipdefault         # (from /etc/ppp/peers/provider)
ipcp-max-failure 30 # (from /etc/ppp/peers/provider)
defaultroute        # (from /etc/ppp/peers/provider)
usepeerdns          # (from /etc/ppp/peers/provider)
noccip              # (from /etc/ppp/peers/provider)
noipx               # (from /etc/ppp/options)
abort on (BUSY)
abort on (NO CARRIER)
abort on (NO DIALTONE)
abort on (ERROR)
abort on (NO ANSWER)
timeout set to 30 seconds
send (AT^M)
expect (OK)
^M
OK
-- got it

```

```

transferpi@raspberrypi: ~ ^ x
Dosya Düzenle Sekmeler Yardım
transferpi@raspberrypi:~$ sudo po
ff
transferpi@raspberrypi:~$ scrot

```

```

expect (OK)
^M
^M
+CPIN: READY^M
^M
OK
-- got it

send (AT+CSQ^M)
expect (OK)
^M
^M
+CSQ: 31,99^M
^M
OK
-- got it

send (AT+CREG?^M)
expect (OK)
^M
^M
+CREG: 0,1^M
^M
OK
-- got it

send (AT+CGREG?^M)
expect (OK)
^M
^M
+CGREG: 0,1^M
^M
OK
-- got it

send (AT+COPS?^M)
expect (OK)
^M
^M
+COPS: 1,0,"BIMcell BIMcell",7^M
^M
OK

```

```

transferpi@raspberrypi: ~ ^ x
Dosya Düzenle Sekmeler Yardım
transferpi@raspberrypi:~$ sudo po
ff
transferpi@raspberrypi:~$ scrot
transferpi@raspberrypi:~$ scrot

```

```

-- got it

send (AT+CGDCONT=1,"IP","internet"^M)
expect (OK)
^M
^M
OK
-- got it

send (ATD*99#^M)
expect (CONNECT)
^M
^M
CONNECT
-- got it

Script chat -s -v -f /etc/chatscripts/chat-connect -T bimcell finished (pid 1203), status = 0x0
Serial connection established.
using channel 1
Using interface ppp0
Connect: ppp0 <--> /dev/ttyUSB3
sent [LCP ConfReq id=0x1 <asynmap 0x0> <magic 0x25357bcd> <pcomp> <accomp>]
rcvd [LCP ConfReq id=0x0 <asynmap 0x0> <auth chap MD5> <magic 0xf2adc5b4> <pcomp> <accomp>]
No auth is possible
sent [LCP ConfRej id=0x0 <auth chap MD5>]
rcvd [LCP ConfAck id=0x1 <asynmap 0x0> <magic 0x25357bcd> <pcomp> <accomp>]
rcvd [LCP ConfReq id=0x1 <asynmap 0x0> <magic 0xf2adc5b4> <pcomp> <accomp>]
sent [LCP ConfAck id=0x1 <asynmap 0x0> <magic 0xf2adc5b4> <pcomp> <accomp>]
sent [LCP EchoReq id=0x0 magic=0x25357bcd]
sent [IPCP ConfReq id=0x1 <addr 0.0.0.0> <ms-dns1 0.0.0.0> <ms-dns2 0.0.0.0>]
sent [IPV6CP ConfReq id=0x1 <addr fe80::7050:df10:08a4:dd54>]
rcvd [LCP DiscReq id=0x2 magic=0xf2adc5b4]
rcvd [LCP EchoRep id=0x0 magic=0xf2adc5b4 25 35 7b cd]
rcvd [IPCP ConfReq id=0x0]
sent [IPCP ConfNak id=0x0 <addr 0.0.0.0>]
rcvd [IPCP ConfNak id=0x1 <addr 10.42.145.212> <ms-dns1 195.175.39.59> <ms-dns2 195.175.39.60>]
sent [IPCP ConfReq id=0x2 <addr 10.42.145.212> <ms-dns1 195.175.39.59> <ms-dns2 195.175.39.60>]
rcvd [IPCP ConfReq id=0x1]
sent [IPCP ConfAck id=0x1]
rcvd [IPCP ConfAck id=0x2 <addr 10.42.145.212> <ms-dns1 195.175.39.59> <ms-dns2 195.175.39.60>]
Could not determine remote IP address: defaulting to 10.64.64.64
Script /etc/ppp/ip-pre-up started (pid 1208)
send (ATD*99#^M)
expect (CONNECT)
^M
^M
CONNECT
-- got it

Script chat -s -v -f /etc/chatscripts/chat-connect -T bimcell finished (pid 1203), status = 0x0
Serial connection established.
using channel 1
Using interface ppp0
Connect: ppp0 <--> /dev/ttyUSB3
sent [LCP ConfReq id=0x1 <asynmap 0x0> <magic 0x25357bcd> <pcomp> <accomp>]
rcvd [LCP ConfReq id=0x0 <asynmap 0x0> <auth chap MD5> <magic 0xf2adc5b4> <pcomp> <accomp>]
No auth is possible
sent [LCP ConfRej id=0x0 <auth chap MD5>]
rcvd [LCP ConfAck id=0x1 <asynmap 0x0> <magic 0x25357bcd> <pcomp> <accomp>]
rcvd [LCP ConfReq id=0x1 <asynmap 0x0> <magic 0xf2adc5b4> <pcomp> <accomp>]
sent [LCP ConfAck id=0x1 <asynmap 0x0> <magic 0xf2adc5b4> <pcomp> <accomp>]
sent [LCP EchoReq id=0x0 magic=0x25357bcd]
sent [IPCP ConfReq id=0x1 <addr 0.0.0.0> <ms-dns1 0.0.0.0> <ms-dns2 0.0.0.0>]
sent [IPV6CP ConfReq id=0x1 <addr fe80::7050:df10:08a4:dd54>]
rcvd [LCP DiscReq id=0x2 magic=0xf2adc5b4]
rcvd [LCP EchoRep id=0x0 magic=0xf2adc5b4 25 35 7b cd]
rcvd [IPCP ConfReq id=0x0]
sent [IPCP ConfNak id=0x0 <addr 0.0.0.0>]
rcvd [IPCP ConfNak id=0x1 <addr 10.42.145.212> <ms-dns1 195.175.39.59> <ms-dns2 195.175.39.60>]
sent [IPCP ConfReq id=0x2 <addr 10.42.145.212> <ms-dns1 195.175.39.59> <ms-dns2 195.175.39.60>]
rcvd [IPCP ConfReq id=0x1]
sent [IPCP ConfAck id=0x1]
rcvd [IPCP ConfAck id=0x2 <addr 10.42.145.212> <ms-dns1 195.175.39.59> <ms-dns2 195.175.39.60>]
Could not determine remote IP address: defaulting to 10.64.64.64
Script /etc/ppp/ip-pre-up started (pid 1208)
Script /etc/ppp/ip-pre-up finished (pid 1208), status = 0x0
not replacing default route to wlan0 [0.0.0.0]
local IP address 10.42.145.212
remote IP address 10.64.64.64
primary DNS address 195.175.39.59
secondary DNS address 195.175.39.60
updetach is set. Now detaching.
transferpi@raspberrypi:~ $

```

The connection was successful.

Possible errors

- Often the connection cannot be identified because the APN is entered incorrectly, and the connection is lost and the modem shuts down.
 - `sudo nano /etc/chatscripts/chat-connect` and correct the APN setting. The required APN can be found online.
- Incorrect entry of the communication port.
 - The necessary application must be reinstalled or re-set with atcom commands.
- Serial port setting is turned off.
 - It is easily activated from the interfaces section of the settings.
- `ttyUSB*` blocked
 - This issue indicates that the communication port is currently busy.
Either close the thing occupying the port or readjust your communication port.

Thanks...

References1: <https://sixfab.com/blog/ppp-installer-for-sixfab-shield-hat/>

References2: <https://docs.sixfab.com/docs/raspberry-pi-3g-4g-lte-base-hat-troubleshooting>

References3: <https://community.sixfab.com/c/raspberry-pi-hats/3g-4g-lte-base-hat/7>

4G Live Streaming

For the project to work, we first need 2 devices with internet access and static ip.

We use motion application here. Open CV can be used in advanced stages.

Of course, these two devices need to be paired first. The first idea is that two devices with static public IPs recognize each other's IP addresses and establish a connection before the connection.



The first phase of the project works in a very similar way to data transfer over wifi;

- Turn on the device to which the camera is connected.
- Install the app.
- Make necessary adjustments and corrections
- Run it.
- And share the publication.

The required application can be "motion or opencv."


```
hasan@raspberrypi:~$ sudo apt-get install motion
Paket listeleri okunuyor... Bitti
Bağımlılık ağacı oluşturuluyor... Bitti
Durum bilgisi okunuyor... Bitti
Aşağıdaki paketler otomatik olarak kurulmuş ve artık bu paketlere gerek duyulmuyor:
  libraspberrypi0 libwpe-1.0-1 libwpebackend-fdo-1.0-1
Bu paketleri kaldırmak için 'sudo apt autoremove' komutunu kullanın.
Aşağıdaki ek paketler kurulacak:
  libmariadb3 libmicrohttpd12 libpq5 mariadb-common mysql-common
Önerilen paketler:
  default-mysql-client postgresql-client
Aşağıdaki YENİ paketler kurulacak:
  libmariadb3 libmicrohttpd12 libpq5 mariadb-common motion mysql-common
0 paket yükseltilecek, 6 yeni paket kurulacak, 0 paket kaldırılacak ve 0 paket yükseltilmeyecek.
979 kB arşiv dosyası indirilecek.
Bu işlem tamamlandıktan sonra 4.394 kB ek disk alanı kullanılacak.
Devam etmek istiyor musunuz? [E/h]
```

```
Bu işlem tamamlandıktan sonra 4.394 kB ek disk alanı kullanılacak.
Devam etmek istiyor musunuz? [E/h] e
İndir: 1 http://deb.debian.org/debian bookworm/main arm64 mysql-common all 5.8+1.1.0 [6.636 B]
İndir: 2 http://deb.debian.org/debian bookworm/main arm64 mariadb-common all 1:10.11.6-0+deb12u1 [24,4 kB]
İndir: 3 http://deb.debian.org/debian bookworm/main arm64 libmariadb3 arm64 1:10.11.6-0+deb12u1 [162 kB]
İndir: 4 http://deb.debian.org/debian bookworm/main arm64 libmicrohttpd12 arm64 0.9.75-6 [112 kB]
İndir: 5 http://deb.debian.org/debian-security bookworm-security/main arm64 libpq5 arm64 15.6-0+deb12u1 [178 kB]
İndir: 6 http://deb.debian.org/debian bookworm/main arm64 motion arm64 4.5.1-2 [495 kB]
11 sn.'de 979 kB alındı (86,9 kB/s)
Paketler önyapılandırılıyor ...
Daha önce seçili olmayan mysql-common paketi seçiliyor.
(Veritabanı okunuyor ... 147649 dosya veya izin kurulu durumda.)
Paket açılacak: ../0-mysql-common_5.8+1.1.0_all.deb ...
Paket açılıyor: mysql-common (5.8+1.1.0) ...
Daha önce seçili olmayan mariadb-common paketi seçiliyor.
Paket açılacak: ../1-mariadb-common_1%3a10.11.6-0+deb12u1_all.deb ...
Paket açılıyor: mariadb-common (1:10.11.6-0+deb12u1) ...
Daha önce seçili olmayan libmariadb3:arm64 paketi seçiliyor.
Paket açılacak: ../2-libmariadb3_1%3a10.11.6-0+deb12u1_arm64.deb ...
Paket açılıyor: libmariadb3:arm64 (1:10.11.6-0+deb12u1) ...
Daha önce seçili olmayan libmicrohttpd12:arm64 paketi seçiliyor.
Paket açılacak: ../3-libmicrohttpd12_0.9.75-6_arm64.deb ...
Paket açılıyor: libmicrohttpd12:arm64 (0.9.75-6) ...
Daha önce seçili olmayan libpq5:arm64 paketi seçiliyor.
Paket açılacak: ../4-libpq5_15.6-0+deb12u1_arm64.deb ...
Paket açılıyor: libpq5:arm64 (15.6-0+deb12u1) ...
Daha önce seçili olmayan motion paketi seçiliyor.
Paket açılacak: ../5-motion_4.5.1-2_arm64.deb ...
Paket açılıyor: motion (4.5.1-2) ...
Ayarlanıyor: mysql-common (5.8+1.1.0) ...
update-alternatives: /etc/mysql/my.cnf (my.cnf) alternatifini sağlaması için /etc/mysql/my.cnf.fallback otomatik kipte kullanılıyor
Ayarlanıyor: libpq5:arm64 (15.6-0+deb12u1) ...
Ayarlanıyor: libmicrohttpd12:arm64 (0.9.75-6) ...
Ayarlanıyor: mariadb-common (1:10.11.6-0+deb12u1) ...
update-alternatives: /etc/mysql/my.cnf (my.cnf) alternatifini sağlaması için /etc/mysql/mariadb.cnf otomatik kipte kullanılıyor
Ayarlanıyor: libmariadb3:arm64 (1:10.11.6-0+deb12u1) ...
Ayarlanıyor: motion (4.5.1-2) ...
Adding group `motion' (GID 122) ...
Done.
adduser: Warning: The home dir /var/lib/motion you specified already exists.
Adding system user `motion' (UID 112) ...
Adding new user `motion' (UID 112) with group `motion' ...
adduser: The home directory `/var/lib/motion' already exists. Not touching this directory.
adduser: Warning: The home directory `/var/lib/motion' does not belong to the user you are currently creating.
Adding user `motion' to group `video' ...
Done.
Created symlink /etc/systemd/system/multi-user.target.wants/motion.service → /lib/systemd/system/motion.service.
Tetikleyiciler işleniyor: man-db (2.11.2-2) ...
```

```
Broadcast message from systemd-journald@raspberrypi (Thu 2024-05-16 23:03:11 +03):
motion[33629]: [0:motion] [EMG] [ALL] motion_startup: Exit motion, cannot create log file /var/log/motion/motion.log: Permission denied

Broadcast message from systemd-journald@raspberrypi (Thu 2024-05-16 23:03:11 +03):
motion[33629]: [0:motion] [EMG] [ALL] motion_startup: Exit motion, cannot create log file /var/log/motion/motion.log: Permission denied

Tetikleyiciler işleniyor: libc-bin (2.36-9+rpt2+deb12u7) ...
hasan@raspberrypi:~$
```

Make any necessary adjustments.

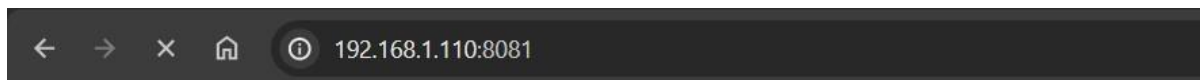

```
hasan@raspberrypi: ~  
GNU nano 7.2 /etc/motion/motion.conf  
# Rename this distribution example file to motion.conf  
#  
# This config file was generated by motion 4.5.1  
# Documentation: /usr/share/doc/motion/motion_guide.html  
#  
# This file contains only the basic configuration options to get a  
# system working. There are many more options available. Please  
# consult the documentation for the complete list of all options.  
#  
#####  
# System control configuration parameters  
#####  
  
# Start in daemon (background) mode and release terminal.  
daemon off  
  
# Start in Setup-Mode, daemon disabled.  
setup_mode off  
  
# File to store the process ID.  
; pid_file value  
  
# File to write logs messages into. If not defined stderr and syslog is used.  
log_file /var/log/motion/motion.log  
  
# Level of log messages [1..9] (EMG, ALR, CRT, ERR, WRN, NTC, INF, DBG, ALL).  
log_level 6  
  
# Target directory for pictures, snapshots and movies  
target_dir /var/lib/motion  
  
# Video device (e.g. /dev/video0) to be used for capturing.  
video_device /dev/video0  
  
# Parameters to control video device. See motion_guide.html  
; video_params value  
  
# The full URL of the network camera stream.  
; netcam_url value  
  
# Name of mmal camera (e.g. vc.ril.camera for pi camera).  
; mmalcam_name value  
  
# Camera control parameters (see raspivid/raspistill tool documentation)  
; mmalcam_params value
```

- `sudo nano` to open the Motion configuration file using the `/etc/motion/motion.conf` command.
- In the file, set the daemon to on (so that it runs in the background).
- Set `webcam_localhost` to OFF (for remote access). Set `stream_localhost` to off (for remote broadcasting).
- Configure camera settings (e.g., width, height, framerate, etc.).

and start the live
broadcast.

```
hasan@raspberrypi:~ $ sudo motion
[0:motion] [NTC] [ALL] conf_load: Processing thread 0 - config file /etc/motion/motion.conf
[0:motion] [NTC] [ALL] mycreate_path: creating directory /var/log/motion
[0:motion] [NTC] [ALL] motion_startup: Logging to file (/var/log/motion/motion.log)
```

Now let's come to connecting to the camera. All we have to do is go to the browser tab and type in the ip address to which the camera is connected.

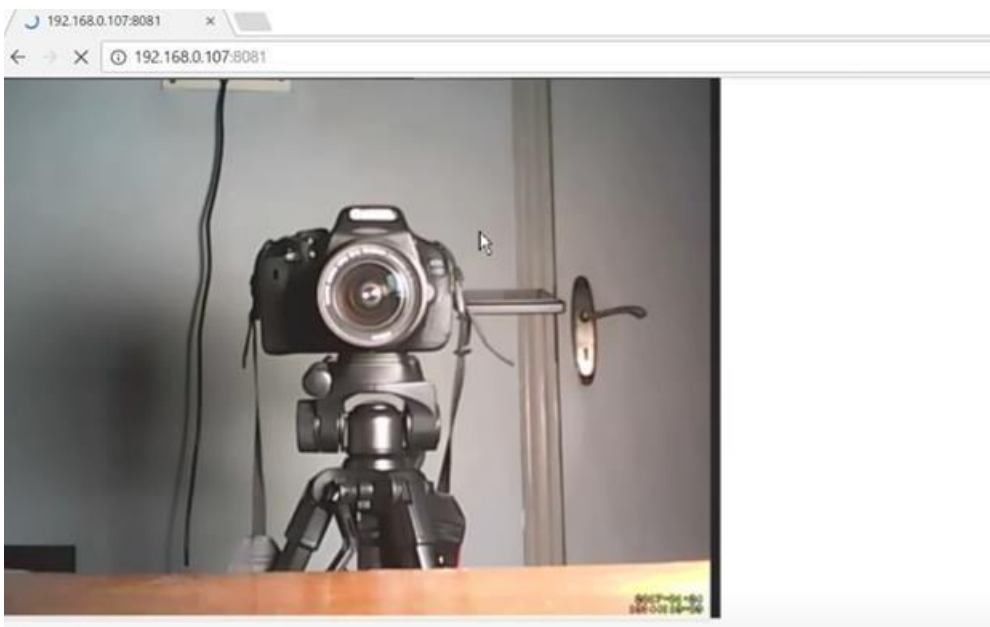


If you noticed that the ip address written here is a local ip, "I hope"

The device broadcasting here is not plugged into a sim card with a static ip, so it shows over the local ip.

If our device was mobile, we would write the static ip address of the camera here.

But normally, it broadcasts directly simultaneously from a fixed location on the internet to the port opened to the remote control in our hand, just like an e-mail.



The rpi4 4gb can do this at an average fps"30-20" at the lowest settings.

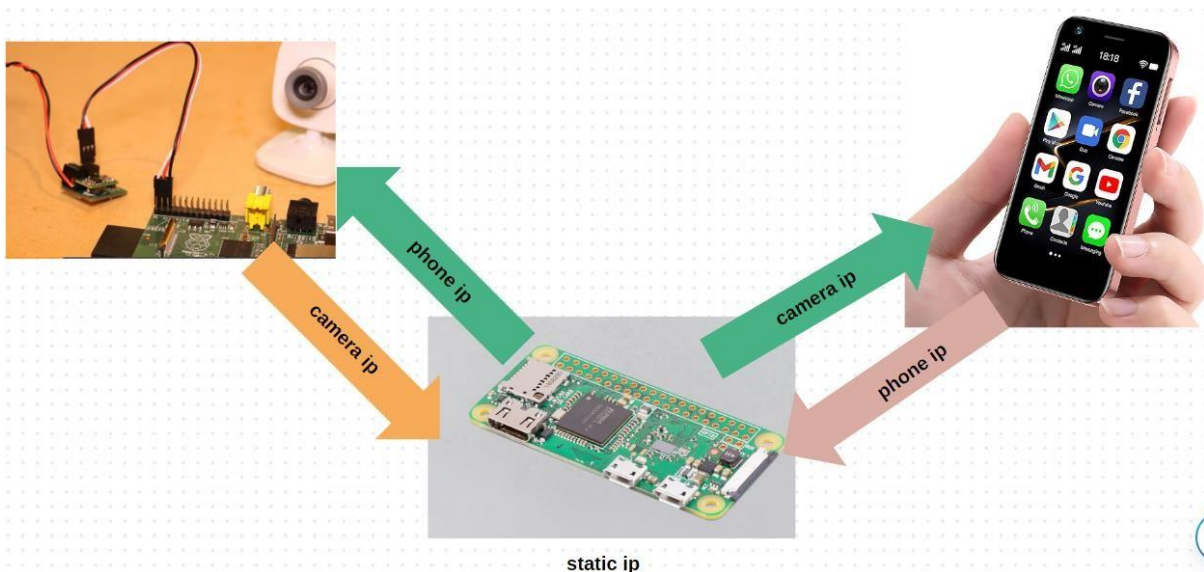


Another idea

Of course, we do not necessarily need sim cards with static IP to carry out this project.

What we need is an intermediate piece of equipment. So a server.

This server has a static ip" it will have a static ip by opening a port over wifi in a fixed location"



- The camera will connect to the server via the internet, and the controller or display owner will be on standby for the device to connect.
- The device with the remote control or screen will still connect to the server and the server will match the ip addresses of these two devices with each other.
- In this way, two sim cards will be able to connect with each other without the m2m "sim card with static ip".
- Here, the server will constantly monitor the IP addresses of the connected devices and ensure that the devices reconnect in case of a change.
- In order to set up this server, certain applications other than opening ports can also be used. For example, cloudflare tunnels.
- We don't need power because the server to do this job has very little work.
- Even one rpi zero can do this matching job.

Arants

- A safer communication in terms of security.
- It can be inspected and easily controlled.
- When necessary, for example, in case of an attack or crash, the system can be shut down via the server.
- lot does not need to buy a line.

Cons

- Possible delay problems as an extra intermediate element is added to communicate.
 - The quality of the work to be done depends on the hardware and internet quality.
 - The system will take on a more sophisticated structure.
 - Since I don't know much about it and it is a completely new idea, it needs to be studied more.
-
- The project is far from being delivered at the moment, image quality, sudden drops in fps and many more that I can't count.
 - Although there is no working prototype yet, it needs to be developed and thought about.
 - I shared with you general information about how the project can be done if you have the necessary equipment.
 - New things can be added all the time. In case the RF signal is lost, the necessary commands can be transmitted directly to the server.

Note: Before reading this report, it is recommended that you read the previous 4G connectivity report. This report assumes that you already have a 4G connection.

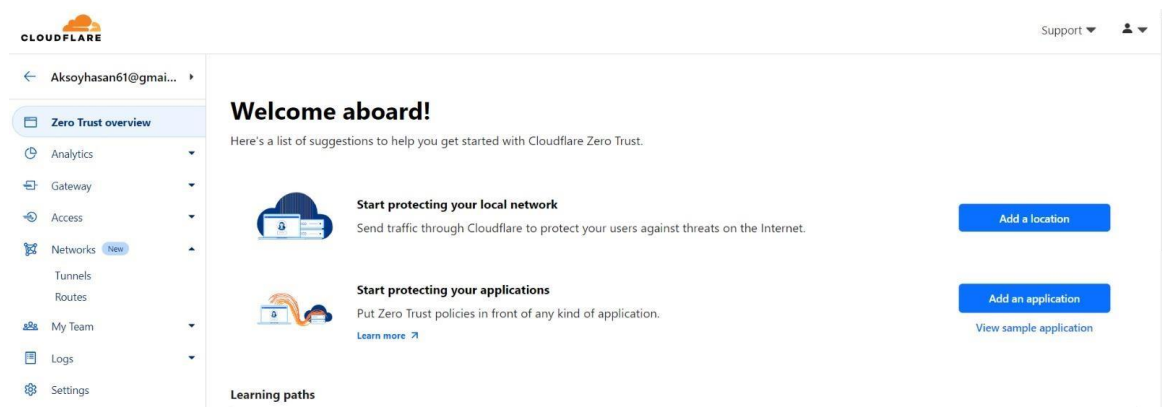
What you need:

- Raspberry Pi 4 (32-bit legacy version)
- 4G connectivity
- 1 camera
- 1 domain and hosting
- Cloudflare account

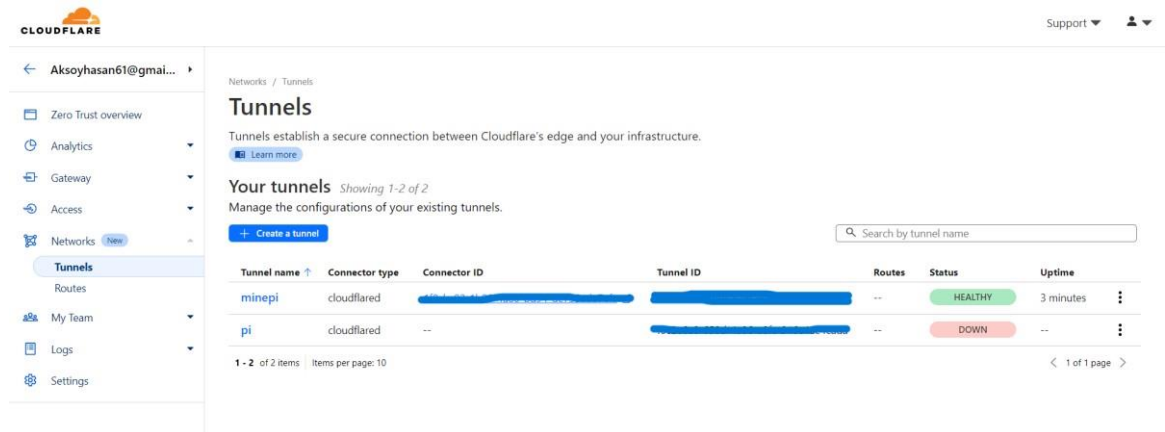
Steps:

- Creating a Cloudflare Account and Adding a Domain • Go to Cloudflare's main site and create an account.
- Add your domain address to Cloudflare.
- Add the Cloudflare DNS provider to the DNS section of your domain.
- Cloudflare Zero Trust and Tunneling

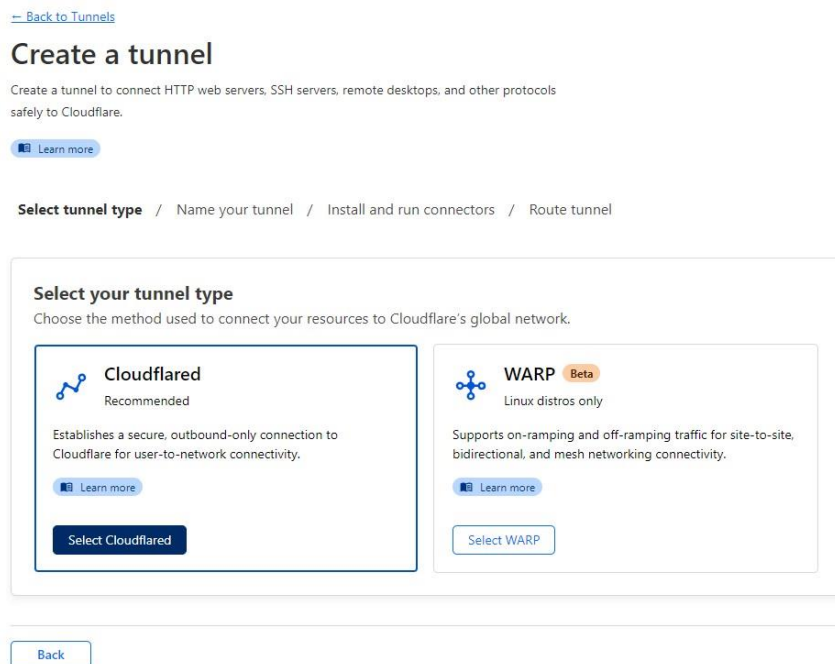
1. Go to the Cloudflare dashboard and click on "Zero Trust".



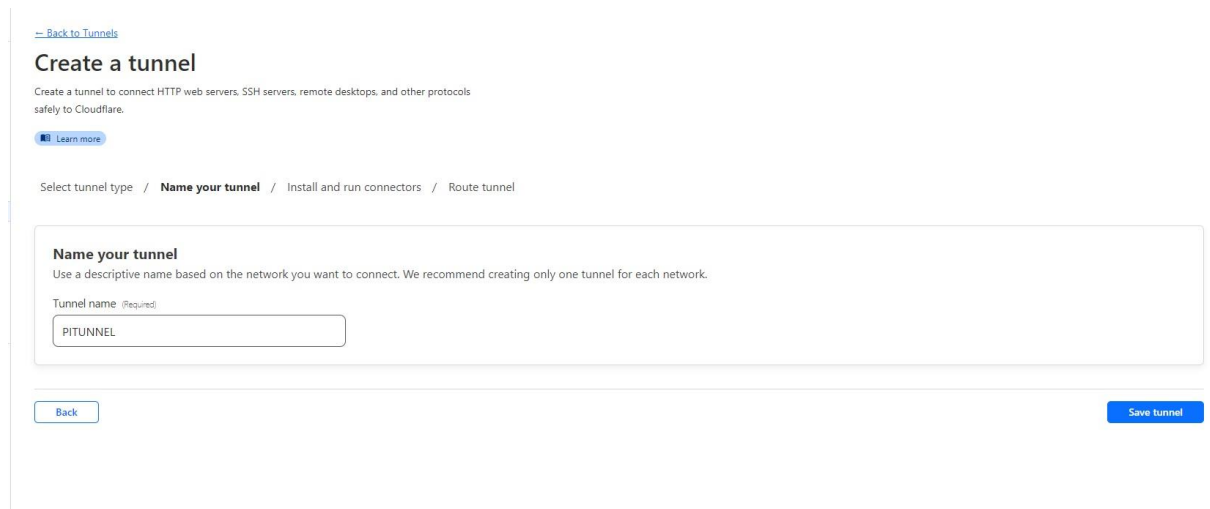
2. From the newly opened dashboard, click on "Networks" > "Tunnels".



3. Select "Create a Tunnel" and select the tunnel type.



4. Give the tunnel a name and move on.



5. In the "Configure" section, select Environment and create a Cloudflare code for the terminal.

Configure

Select tunnel type / Name your tunnel / **Install and run connectors** / Route tunnel

Choose your environment

Choose an operating system:

Windows Mac Debian Red Hat Docker

Choose an architecture:

64-bit 32-bit arm64-bit arm32-bit

Install and run a connector

To connect your tunnel to Cloudflare, copy-paste one of the following commands into a terminal window. Remotely managed tunnels require that you install cloudflared 2022.03.04 or later.

ⓘ **Store your token carefully.** This command includes a sensitive token that allows the connector to run. Anyone with access to this token will be able to run the tunnel. ✕

If you don't have cloudflared installed on your machine:

```
curl -L --output cloudflared.deb https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-linux-arm64.deb &&
sudo dpkg -i cloudflared.deb &&
sudo cloudflared service install eyJhIjo1ZD...
```

If you already have cloudflared installed on your machine:

```
sudo cloudflared
$ service install
eyJhIjo1ZD...
```

6. Since the rasberian version for Raspberry Pi 4 is not here, you will only use the code on the right that allows you to set up the tunnel instead of the code on the left for downloading Cloudflare. Don't lose this code, it will be needed in the future.

NOTE: If you have lost the code, you can get the code again from the "Configure" option by clicking the three dots next to the "inactive" text in the tunnels section.

7. Specify the domain name to connect the tunnel to. For example: (pitunnel.domeinname z.com)

[← Back to Tunnels](#)

Route Traffic

Select tunnel type / Name your tunnel / Install and run connectors / **Route tunnel**

💡 Route traffic by adding a public hostname or a private network to your tunnel. You can always add more hostnames or networks at a later time.

🔒 Protect your resource by adding an Access policy under [Access > Applications > Self-hosted](#).

Public Hostnames Private Networks

Add public hostname for PITUNNEL

Public hostname

Subdomain	Domain (Required)	Path (optional)
pitunnel	example.com	

Service

Type (Required)	URL (Required)
HTTP	localhost:5000

For example, <https://localhost:5001>

[Additional application settings](#)

8. In the service section, select the "http localhost:5000" option.

Service

Type (Required)	URL (Required)
HTTP	localhost:5000

For example, <https://localhost:6001>

9. There remains one last step for installation. On the Raspberry Pi 4, download Cloudflare and write the code previously received in the terminal. These posts should appear

```
2023-01-21T22:00:33Z INF Using Systemd
2023-01-21T22:00:36Z INF Linux service for cloudflared installed successfully
```

10. The connection must be established successfully. Go to the tunnels section again and check the tunnel. The green "HEALTHY" mark indicates that the tunnel has been successfully installed.

Tunnels establish a secure connection between Cloudflare's edge and your infrastructure.

[Learn more](#)

[Create a tunnel](#)

Tunnel name	Status	Routes	Uptime	Created
▼ PiTunnel 11111111-dddd-4444-8888-ddddd123 bdd2	HEALTHY	--	32 minutes	21 January 2023

[Configure](#) | [Delete](#)

Camera Streaming with Raspberry Pi 4

Flask is used for camera streaming on Raspberry Pi 4.

To run the Flask code, you need to download the OpenCV library.

Alternatively, Motion or GStreamer can be preferred.

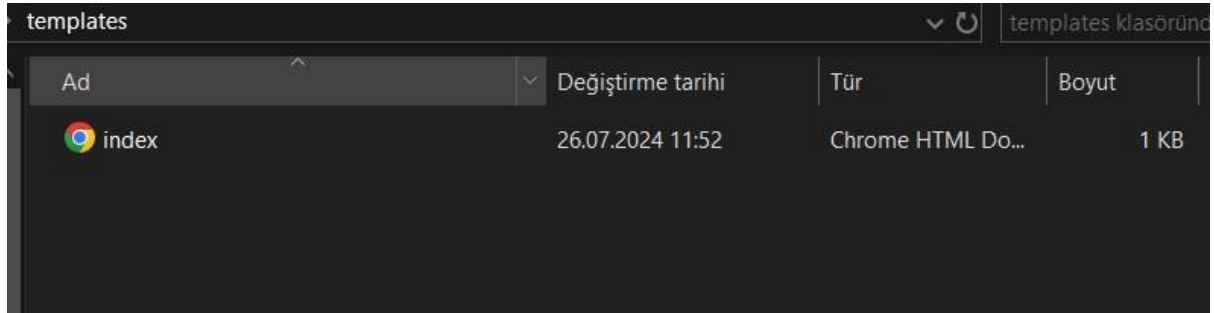
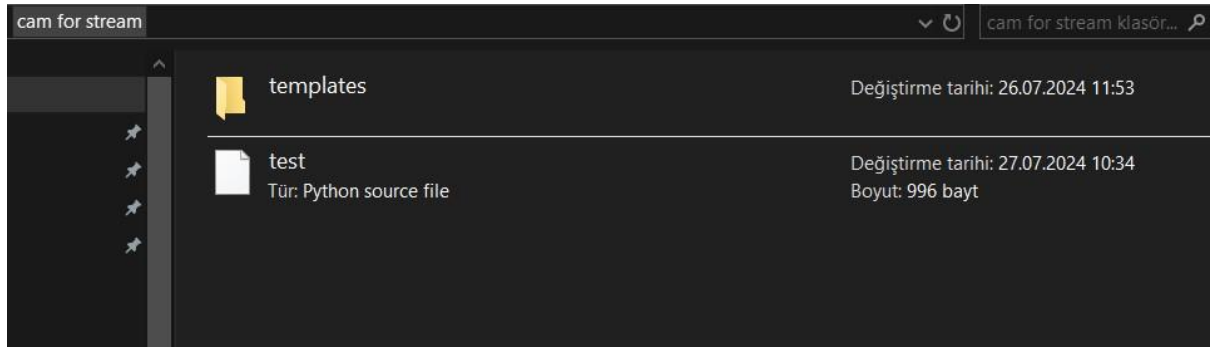
Save the code and run it on the Raspberry Pi

```

1  √ from flask import Flask, render_template, Response
2  | import cv2
3
4  app = Flask(__name__)
5
6  √ def gen_frames():
7  |     cap = cv2.VideoCapture(0) # Kamerayı başlat
8  |     if not cap.isOpened():
9  |         print("Kamera açılmadı!")
10 |         exit()
11
12 |     while True:
13 |         ret, frame = cap.read()
14 |         if not ret:
15 |             print("Kare okunamadı!")
16 |             break
17
18 |         # JPEG formatında görüntüyü sıkıştır
19 |         ret, buffer = cv2.imencode('.jpg', frame)
20 |         frame = buffer.tobytes()
21
22 |         # JPEG formatında görüntüyü HTTP yanıtı olarak gönder
23 |         yield (b'--frame\r\n'
24 |               b'Content-Type: image/jpeg\r\n\r\n' + frame + b'\r\n')
25
26 |     cap.release()
27
28 @app.route('/')
29 √ def index():
30 |     return render_template('index.html')
31
32 @app.route('/video_feed')
33 √ def video_feed():
34 |     return Response(gen_frames(), mimetype='multipart/x-mixed-replace; boundary=frame')
35
36 √ if __name__ == '__main__':
37 |     app.run(host='0.0.0.0', port=5000)
38

```

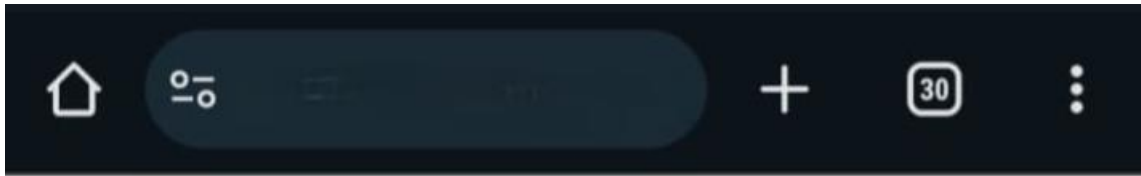
NOTE: the code will not work without the index.html file inside the template file in the same folder.



index.html content is as follows.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Kamera Görüntüsü</title>
</head>
<body>
  <h1>Kamera Görüntüsü</h1>
  
</body>
</html>
```

Image:



Kamera Görüntüsü



Conclusion:

This report details the steps to be followed to perform camera streaming on the Raspberry Pi 4 using the 4G connection. The step-by-step process of creating a secure tunnel using Cloudflare, connecting through it, and streaming cameras with Flask is explained.

The successful implementation of this project, together with the effective use of 4G connectivity and the security solutions offered by Cloudflare, makes it possible to create a low-cost and efficient remote monitoring system.